

Multiple Choice

1. **Answer:** C
Options: A) 1; B) 2; C) 3; D) 4
Note: Correct option: C - 3
2. **Answer:** D
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: D - False, True
3. **Answer:** B
Options: A) 0; B) 2; C) 1; D) 3
Note: Correct option: B - 2
4. **Answer:** B
Options: A) 6; B) 12; C) 30; D) 105
Note: Correct option: B - 12
5. **Answer:** B
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: B - False, False

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '(1,6)'.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '6'.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** We consider the opposite; we try to find the number of words that do not contain A, and then subtract it from the total possible number of words. So we have a few cases to consider: • One letter words: There is only 1 one-letter word that contains A, that's A. • Two letter words: There are $19 \times 19 = 361$ words that do not contain A. There is a total of $20 \times 20 = 400$ words, so we have $400 - 361 = 39$ words that satisfy the condition. • Three letter words: There are $19 \times 19 \times 19 = 6859$ words without A, and there are $20^3 = 8000$ words available. So there are $8000 - 6859 = 1141$ words that satisfy the condition. • Four letter words: Using the same idea as above, we have $20^4 - 19^4 = 29679$ words satisfying the requirement. So this gives a total of $1 + 39 + 1141 + 29679 = 30860$ words.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** True, True. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.